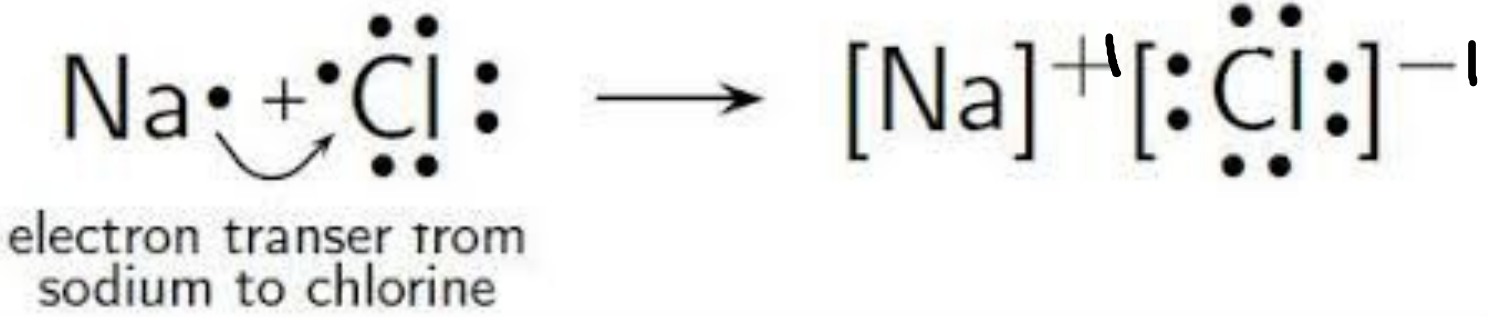
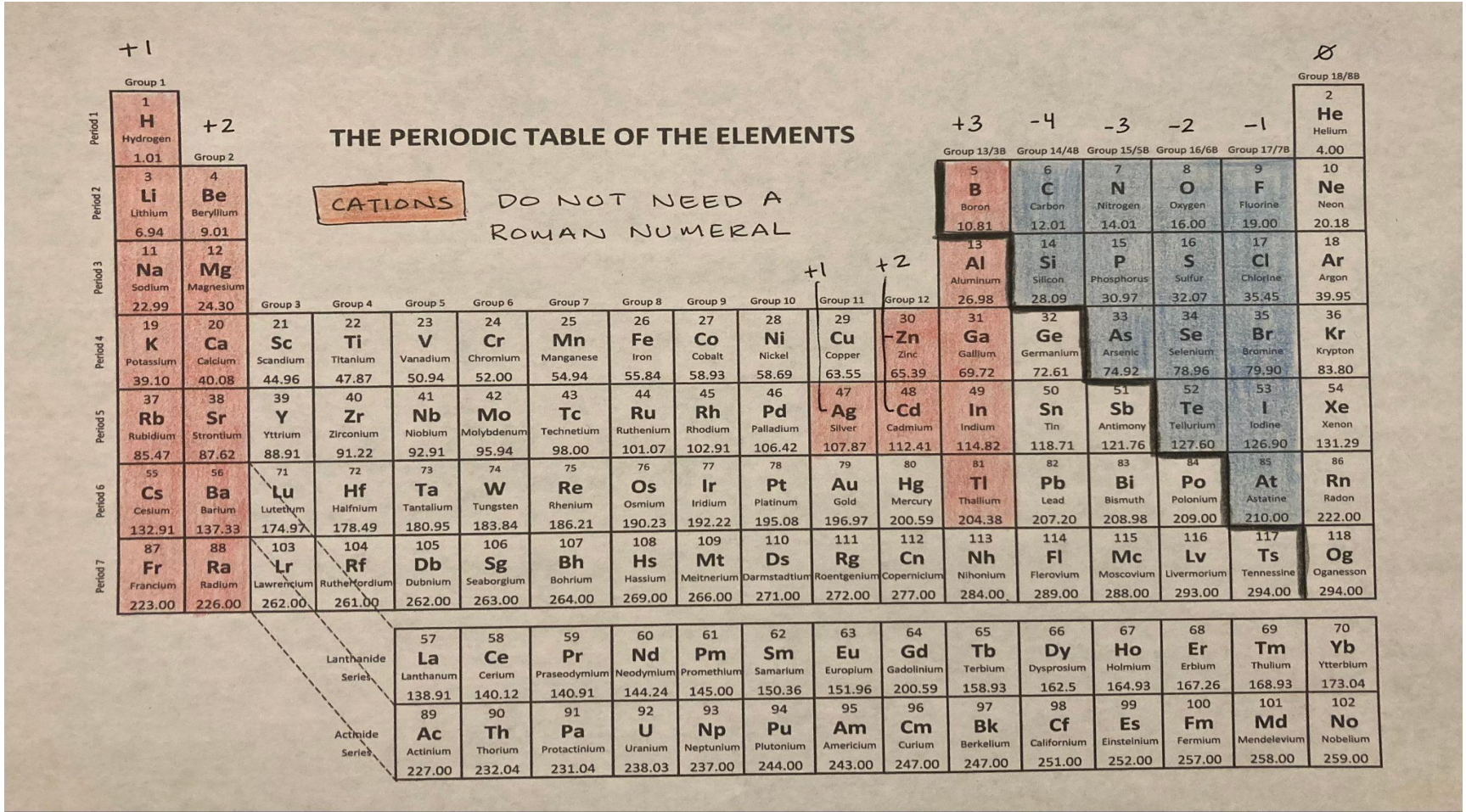
Ionic Bonding - Type I

Ionic Compound

- Ionic - involves the transferring of electron(s)
  - Cation - “+” ion formed when an atom loses electron(s)
  - Anion - “-” ion formed when an atom gains electron(s)



Ion Charges - Type I



Type I Nomenclature

Red + Blue

- Type I Binary
  - Compound formed from a metal (1A, 2A, 3A, and Zn, Cd, or Ag) + nonmetal
- Ionic Formulas

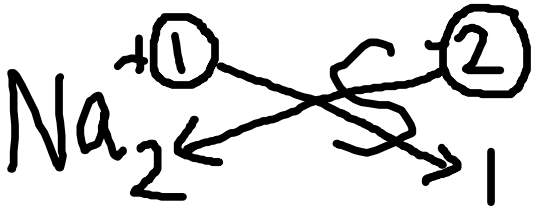


  - Write each ion, cation first. Do not show charges in the final formula.
  - Overall charge must be zero (neutral)
    - If charges cancel, just write the symbols.
    - If not, use subscripts to balance the charges (aka criss-cross)

Type I Nomenclature

Na^{+1} Cl^{-1} NaCl

- Writing Type I Binary Ionic Formulas is simple. If the charges of the ions do not balance each other out, use the criss-cross.
- Example: Na^{+1} and S^{-2} (these charges do not cancel each other out)



Na_2S

+2 -2

- When the charges cancel, the formula is complete.
- You must reduce the subscripts if needed!

Type I Nomenclature

- Ionic Names
  - Write the name of the cation (+) as it's found on the periodic table.
  - Write the name of the anion (-). Be sure to change the ending to -ide.
- Example: Ca_3N_2

calcium nitride

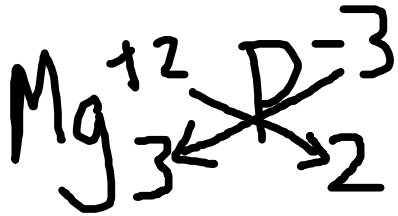
Type I Nomenclature

Potassium chloride



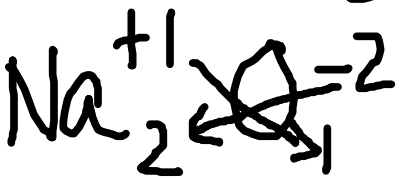
KCl

Magnesium phosphide



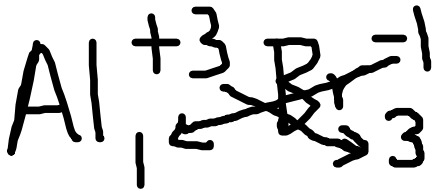
Mg_3P_2

Sodium oxide



Na_2O

Aluminum bromide



$AlBr_3$

Silver sulfide



Ag_2S

Type I Nomenclature

$AlBr_3$ aluminum bromide

Li_3N lithium nitride

ZnF_2 zinc fluoride

CaO calcium oxide

Cd_3P_2 cadmium phosphide